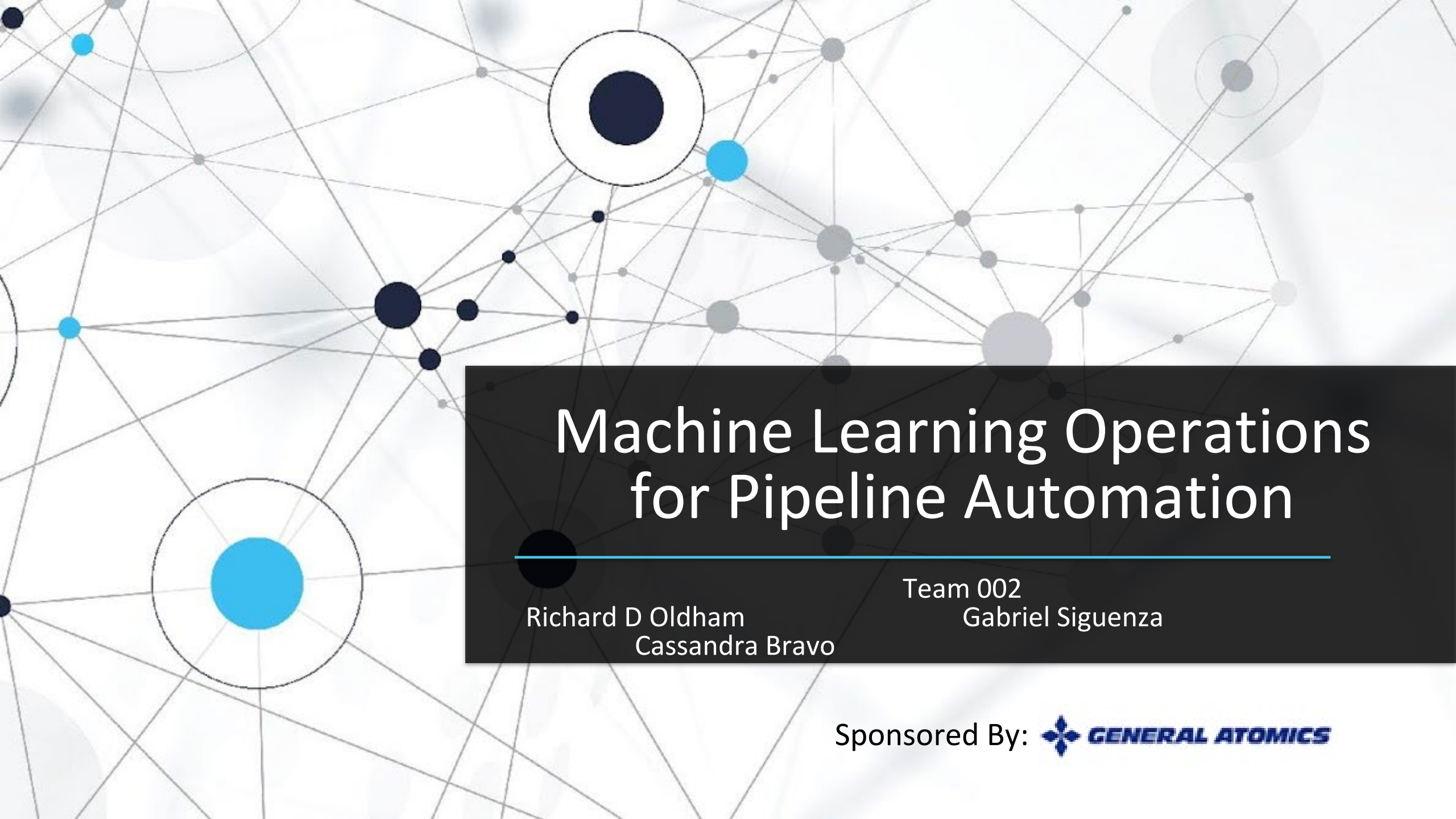




Machine Learning Operations for Pipeline Automation

Richard D Oldham
Cassandra Bravo

Team 002
Gabriel Siguenza

Sponsored By:  **GENERAL ATOMICS**

Special Thanks!

Fall Industry Mentor

Teresa Westfall



Spring Industry Mentor

Hernan Silva-Galvan



PRESENTATION OUTLINE

- ❖ **Problem Statement** – What and Why
- ❖ **Project Design** – System/Pipeline Design
- ❖ **Project Implementation** – Programming languages/skills used, testing techniques, tools, video demo, summary table, user requirements review
- ❖ **Lessons Learned**
- ❖ **To the next class of students..**

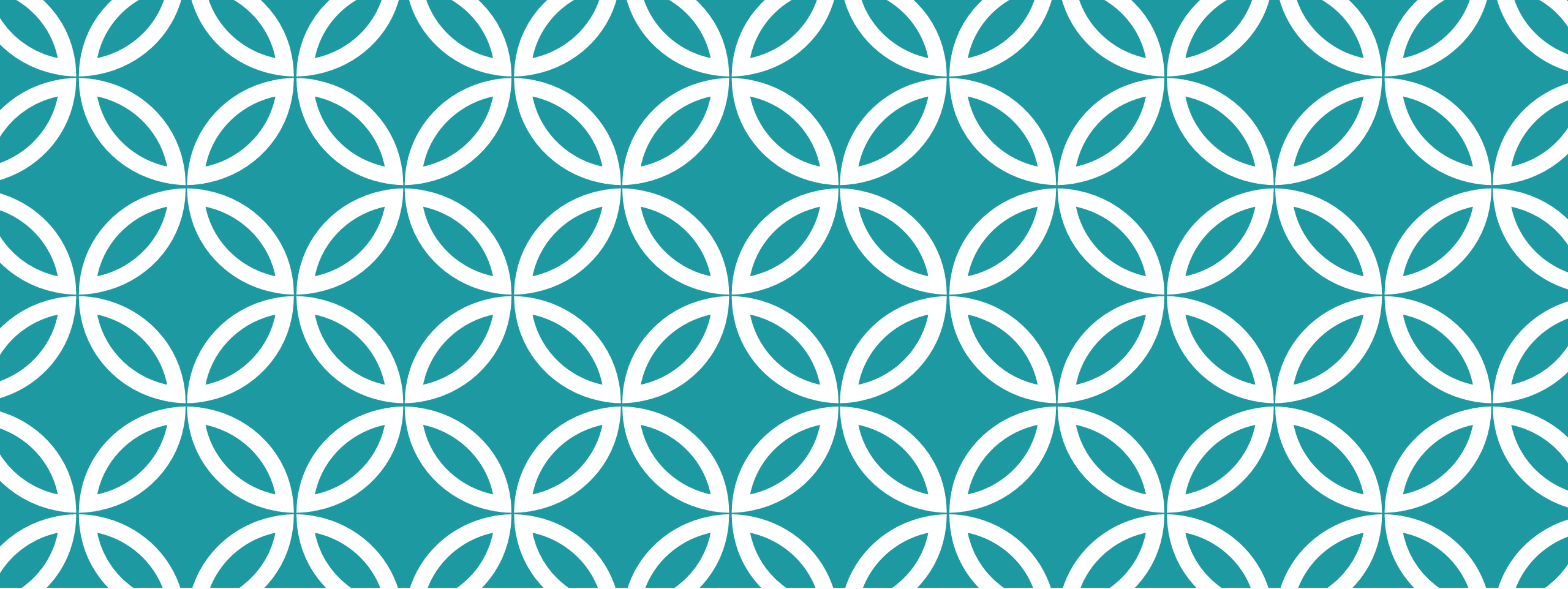
PROBLEM STATEMENT

What is this project?

- 2 Machine Learning models working together to classify images with high accuracy
- One model (GAN) creates fake data and is assigned a label based on class it is derived from
- Fake data is then fed to the other model (CNN) for classification
- Best model is picked based on performance

Why do we need it?

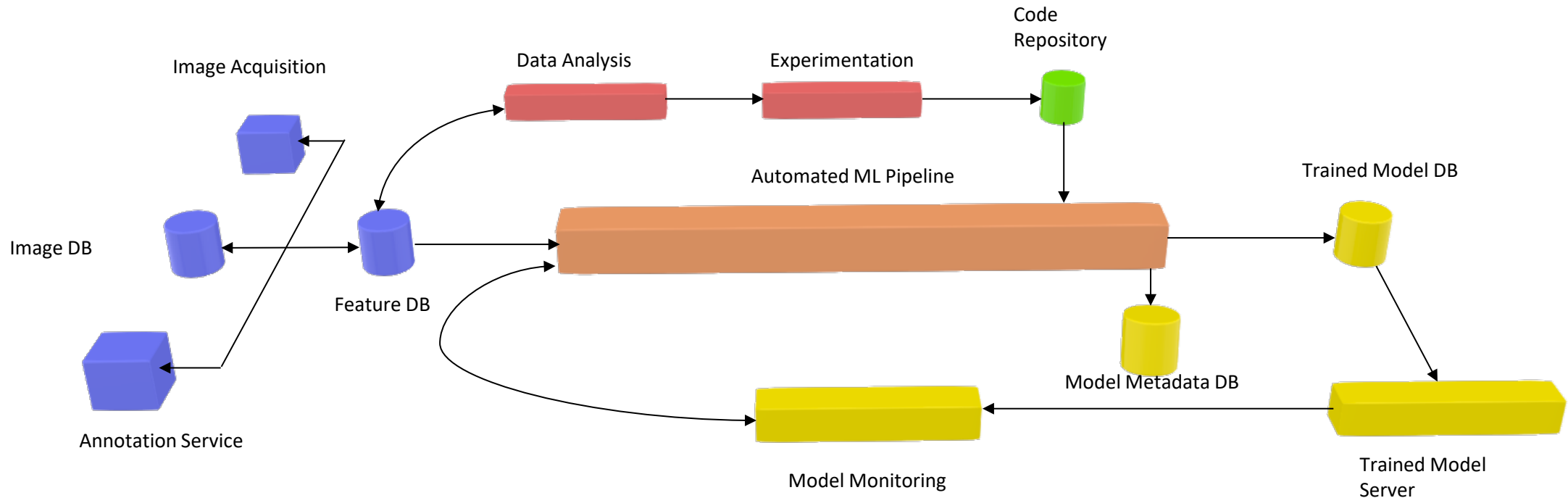
- Data Augmentation
- Too much human interference—make the computer do it for us!
- Save money, time, resources, money
- Keep humans safe on the ground



PROJECT DESIGN

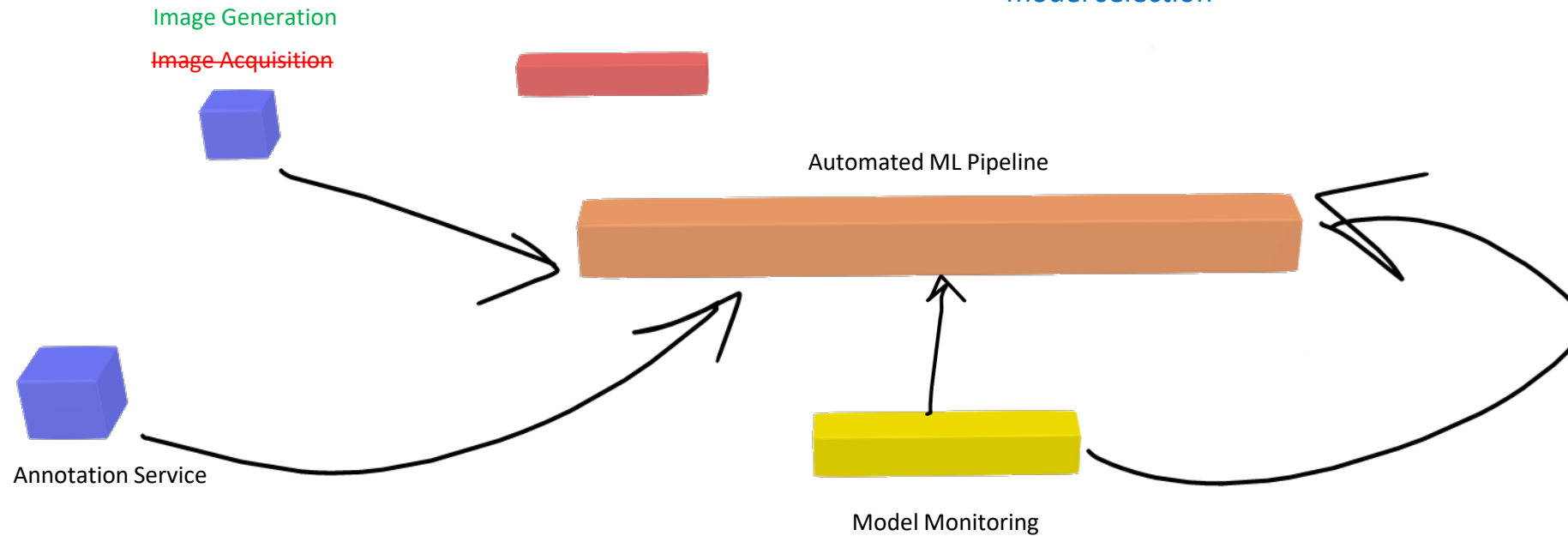


TYPICAL IMAGE PROCESSING MACHINE LEARNING (ML) SYSTEM



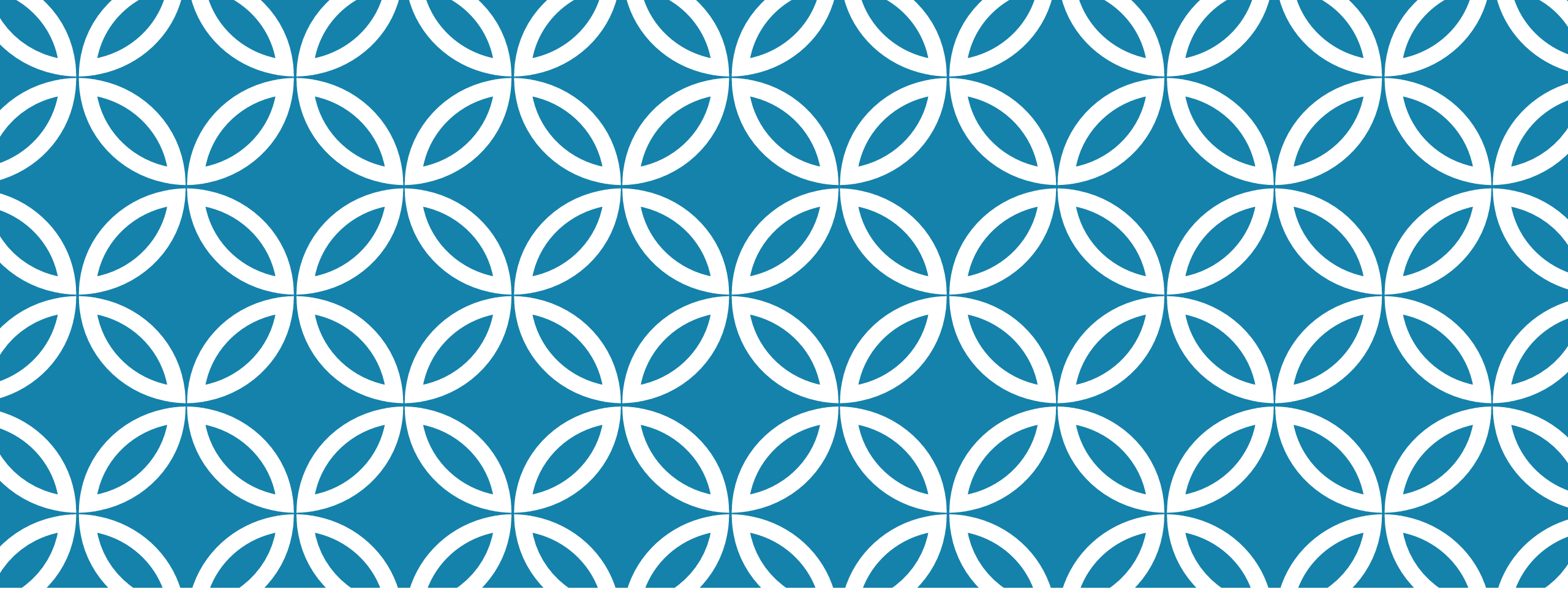
A DIFFERENT APPROACH

- integration and deployment of the ideal AI models
- model performance report generation
- retraining of models with new data
- training data labeling
- model selection



OUR SYSTEM





PROJECT IMPLEMENTATION

PROGRAMMING LANGUAGE, SKILLS USED

Pytho
n



Why?

Ease of Learning and
Use

Strong Support for
Scientific Computing

Extensive Machine
Learning Libraries

“The best way to learn a language is to
speak to natives.”

The guy learning python:



TESTING TECHNIQUES, TOOLS



Python Unit Tests

[Demo](#)



PyTest



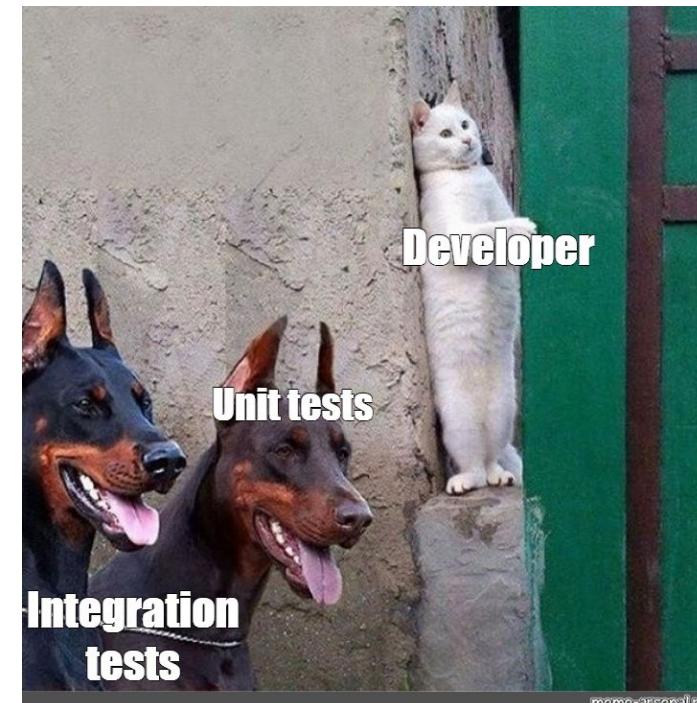
Continuous Integration



Automated Testing through GitLab



Tensor Board and Weights & Biases



Demo 1: CNN, 320 training images



Demo 2: GAN fake image generation



Demo 3: CNN, 320 + 640 fakes





SUMMARY TABLE

User Requirements	Completed?
The user should be able to use external tools and frameworks	Yes
The user should be able to create an optimized model	Yes
The user should be able to generate a report	Yes

What Did We Learn?

- ❖ AI/ML is difficult..but NOT impossible!
- ❖ Rabbit holes can be fun (if they are the right ones to fall down into)
- ❖ Environment build & configuration is crucial
- ❖ Software is definitely not one-size-fits-all
- ❖ There are just some things TDD can't be used for. This is one of those things.

What Did We Learn?

On Machine Learning

Understand the Problem

Iterative Process

Keep Up-to-Date

Operationalizing the Model

On Agile

Embrace Change

Incremental Progress

Regular Reflection and Adjustment

Friend: *Loosing his mind debugging his tensorflow code*
Me, an intellectual:



Closing Words of Wisdom...

- ❖ Ask questions often and early!!
- ❖ If something seems like too much at once, break it down into digestible pieces
- ❖ Ask for help early—don't take too long trying to figure it out yourself; You will lose precious time this way
- ❖ Meet as often as possible, with each other, with IM/FA
- ❖ Use ALL of your resources, including faculty you've never spoken to before (no time to be shy!)
- ❖ Trust in yourself, and in your team :)



Closing Words of Wisdom...

Diversify your experience

“Git” into personal projects, motivate yourself to showcase your effort through GitHub

Learn more languages

Play with anything open source!

Know limitations on performance and understand complexities early